SHASMITHA JEGADHEESAN

ASPIRING INFORMATION SECURITY ENTHUSIAST

@Shasmitha Jegadheesan

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TryHackMe: @VelvetShark
Portfolio: Shasmitha Jegadheesan

Detail-oriented Cybersecurity undergraduate with a strong foundation in network analysis, vulnerability management, and defensive security frameworks. Hands-on experience designing secure environments and evaluating access controls through virtual labs and practical tools (Wireshark, Nmap, Nessus, Burp Suite). Knowledgeable in industry standards including NIST CSF, ISO/IEC 27001, and PCI DSS. A clear communicator eager to leverage technical risk assessment skills to support and safeguard organizational infrastructure.

TECHNICAL SKILLS

SECURITY TOOLS:

Wireshark (Network Traffic Auditing), Nmap (Asset Discovery), Nessus/Qualys (Vulnerability Assessment), Metasploit, Burp Suite

GRC & COMPLIANCE FRAMEWORKS:

NIST CSF, PCI DSS, SOX (Familiarity), ISO/IEC 27001 Foundation.

SPECIALIZATIONS:

Technical Risk Assessment, Vulnerability Management, Policy & Documentation, Access Control Validation, Auditing & Logging.

ENVIRONMENTS & PLATFORMS:

Virtualization:

- Oracle Virtual Box, VMWare Workstation Pro 25H2

Security, O/S & Training Platforms:

- TryHackMe, Hack The Box, Metasploitable2, GitHub, OWASP top 10, Parrot OS, Kali Linux, Ubuntu

Languages:

- English (Fluent), Sinhala (Fluent), Tamil (Fluent).

TECHNICAL PROJECTS & ACHIEVEMENTS

ADVANCED SECURITY LAB: VULNERABILITY ASSESSMENT & CONTROL VERIFICATION

- **Risk Identification:** Utilized virtualized infrastructure (Kali Linux and Metasploitable2 target systems) to conduct systematic technical risk assessments and discover misconfigurations.
- **Vulnerability Mapping:** Evaluated and documented system exposures, mapping identified vulnerabilities against common compliance guidelines (OWASP Top 10) to propose remediation steps.

SECURE MULTIPROCESSOR TCP SERVER: CUSTOM PROTOCOL, IAM & AUDIT LOGGING

- **Control Design & Implementation:** Architected a secure, concurrent C/Python application featuring custom session management and strict Identity & Access Management (IAM) controls.
- **Brute-Force & Access Controls:** Implemented an automated account lockout mechanism (preventing dictionary/brute-force attacks) following three consecutive failed login attempts, demonstrating real-world implementation of NIST access control guidelines.
- **Security Auditing & Logging:** Developed a robust, persistent logging system to record chronological security events including client IPs, process IDs, executed commands, and system responses to ensure absolute audit traceability.
- **Resource Integrity:** Programmed signal handlers (handling child processes via waitpid to prevent zombie processes) to maintain system availability and performance baselines.

COMPETITIVE CAPTURE THE FLAG (CTF) EXPERIENCE

- **ICTS CTF (2026):** Secured 25th place out of 76 teams, advancing to Phase 3. Applied skills in OSINT, web penetration testing, and reverse engineering.
- **Battle of the Multiverse '26 (International):** Participated in a 24-hour challenge covering malware analysis, digital forensics, and full-pwn scenarios.
- **TryHackMe Performance:** Achieved 1st Place in the Gold League (following a 1st Place win in the Silver League), demonstrating persistent improvement in offensive security labs.

ADVANCED DECEPTION TECHNOLOGY (ADT) RESEARCH & LAB DESIGN

- **Architected Deceptive Architectures:** Researched and designed the strategic deployment of advanced active defense mechanisms, including honeypots, honey-files, and honeytokens, to misdirect attackers and detect internal network lateral movement.
- **Threat Detection & Mapping:** Analyzed the evolution of deception technology spectrums to understand how proactive defense mechanisms enhance real-time incident response and early-stage threat identification within a SOC framework.

VELVET SHARK STUDIO | CYBERSECURITY PORTFOLIO AND BLOG

- Publishes technical write-ups and step-by-step breakdowns of CTF challenges and lab configurations.
- Translates complex vulnerabilities into actionable documentation, leveraging skills in technical writing and scriptwriting.

EDUCATION

BSC (HONS) IN INFORMATION TECHNOLOGY (SPECIALIZING IN CYBERSECURITY)

Sri Lanka Institute of Information Technology (SLIIT) Kandy UNI
Current Status: 3rd Year, 1st Semester | Expected Graduation: 2028 (IT24104218)

SECONDARY EDUCATION

K/Madeena National College: GCE Advanced Level (2021 - 2023)
Hill Country International Girls College: GCE Ordinary Level (2009 - 2020)

CERTIFICATIONS

ISC2:
- Certified in Cybersecurity (CC) Candidate

Qualys:
- Vulnerability Management Foundations.

TryHackMe:
Pre-Security Path Completion, Love at First Breach Completion.

Cisco:
- Introduction to Cybersecurity.

SysCare IT Solutions:
- Cybersecurity Webinar (Ethical Hacking Fundamentals).

Palo Alto Networks:
- Security Operations Fundamentals

SOFT SKILLS & LEADERSHIP

Talent management: Active member of AIESEC in SLIIT Kandy UNI (Outgoing Global Talent department) since March 2026, Team Leader for OGT CXP since July 2026

Leadership: Served as Senior Prefect at Madina National School; managed student discipline and large-scale event logistics.

Communication: Head of Script Writing for the Media Unit and Sports Union Co-ordinator; experienced in public speaking and technical documentation.

Quick Thinking & Problem Solving: Proven ability to remain calm and analytical under pressure during 24-hour CTF competitions.

CERTIFIED ACCURATE AS OF AUGUST 2026.